

# Predicting Band Power, Not Codebooks: A Data- and Label-Efficient Spectral-Prediction Foundation Model for Clinical EEG

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## Abstract

Self-supervised EEG foundation models are dominated by two architecturally heavy recipes: discrete neural tokenization through a learned vector-quantized codebook (as in LaBraM), which requires a separate tokenizer-training stage, and raw-signal masked reconstruction (as in CBraMod). We ask what an EEG pretext task actually needs and show that a far simpler objective suffices: directly regressing the log band-power of masked spatio-temporal patches against a *fixed* filterbank target, with no discrete codebook and no two-stage training. Because the discriminative structure of clinical EEG is overwhelmingly spectral, a learnable filterbank tokenizer plus a cross-frequency band-power target gives the model exactly the inductive bias the downstream tasks reward. Pretrained on only  $\sim 2,500$  hours of EEG—a random one-seventh subset of the TUH EEG Corpus, comparable to the smallest prior foundation model and roughly a quarter of CBraMod’s data—our model reaches state-of-the-art sleep staging (IS-RUC, HMC) and stays competitive across clinical spectral tasks. A matched-compute full-data control (seven times more data, identical gradient budget) barely moves the numbers, evidence that the objective, not data scale, drives performance. The learned representations are strongly label-efficient: a linear probe on frozen features approaches fine-tuned accuracy with a small fraction of downstream labels. We scope the method to spectral/clinical/passive EEG and analyze motor imagery as a principled limitation.

## Introduction

Large models pretrained on the TUH EEG Corpus (TUEG) have become the standard recipe for EEG representation learning (Jiang et al. 2024; Wang et al. 2025). The two dominant designs are both heavy. LaBraM (Jiang et al. 2024) first trains a vector-quantized neural tokenizer and then predicts its discrete codes, a two-stage pipeline whose first stage is itself a nontrivial spectral autoencoder. CBraMod (Wang et al. 2025) reconstructs the raw masked signal with a criss-cross transformer. Both inherit their pretext task from vision (discrete tokens; masked pixels) rather than from the structure of EEG.

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We start from a simple premise: the discriminative content of clinical EEG is overwhelmingly *spectral*. Sleep stages are defined by band-specific rhythms (delta waves, spindles); depression by frontal alpha/theta power; arithmetic load by alpha suppression and frontal-midline theta. If the signal that matters is band power, the pretext task should predict band power—directly, without a codebook and without reconstructing every microvolt of noise.

We introduce a **codebook-free spectral-prediction** foundation model. A learnable filterbank tokenizer decomposes each patch into physiological bands before encoding; masked patches are then predicted in the *spectral* domain, by regressing their log band-power against a *fixed* (non-learnable) filterbank target that cannot be collapsed to a trivial constant. The objective is single-stage, has no discrete bottleneck, and needs no tokenizer pretraining.

Beyond simplicity, our central empirical finding is about *efficiency*. Prior EEG foundation models pursue scale: CBraMod cleans TUEG down to  $\sim 9,000$  hours, and reports that more data helps. We find the opposite for spectral tasks. Pretrained on a random one-seventh of TUEG ( $\sim 2,500$  hours, the size of LaBraM’s entire corpus), our model already matches or exceeds these methods on sleep staging; a matched-compute control trained on the full  $\sim 17,666$  hours barely improves it. Data scale is not the bottleneck; the objective is.

Our contributions:

- **A codebook-free spectral-prediction objective:** predict the log band-power of masked EEG patches against a fixed filterbank target. Single-stage, no vector quantization, no reconstruction of raw noise. A fixed target is necessary—a learnable target collapses.
- **Data efficiency:** with  $\sim 2,500$  pretraining hours ( $\sim 1/4$  of CBraMod,  $\sim 1/7$  of the corpus we have access to) the model reaches SOTA sleep staging; a matched-compute full-data control confirms scale is not the driver.
- **Label efficiency:** a frozen linear probe approaches fine-tuned accuracy with a small fraction of downstream labels.
- **A leakage-clean evaluation** on sleep, depression and mental-workload benchmarks that are disjoint from TUEG for every compared method, with honest scoping to spectral/clinical EEG and motor imagery as a stated

limitation.

## Related Work

**EEG foundation models.** LaBraM (Jiang et al. 2024) learns a neural vector-quantized tokenizer and predicts discrete codes; CBraMod (Wang et al. 2025) pairs criss-cross attention with masked signal reconstruction; CSBrain (Chen et al. 2025) adds cross-scale spatio-temporal modeling. All predict either a discrete token or the raw signal. We instead regress a continuous *spectral* target (band power) with no codebook and no reconstruction head as the primary objective.

**Spectral targets in SSL.** Predicting spectral energy is used in speech and audio (log-mel reconstruction) and appears implicitly inside LaBraM’s VQ tokenizer, which is trained to reconstruct Fourier amplitude/phase. We make the spectral target the *explicit, primary, codebook-free* pretext task for the foundation model itself, and show a fixed (non-learnable) filterbank target is required to avoid collapse.

**Data scale.** CBraMod reports that scaling cleaned TUEG toward  $\sim 9,000$  hours improves downstream accuracy. LaBraM curates a diverse  $\sim 2,500$ -hour corpus. Our data-efficiency study reaches the same conclusion from the other direction: on spectral tasks, a random  $\sim 2,500$ -hour subset matches the full corpus at equal compute.

**Benchmark leakage.** TUEG-derived pretraining overlaps at the patient level with the TUH downstream sets (TUAB/TUEV). Our benchmarks—sleep (ISRUC, HMC), depression (Mumtaz), mental arithmetic—are disjoint from TUEG for *every* method and thus directly comparable; we use the same subject-disjoint splits as CBraMod/CSBrain.

## Method

**Input.** A 10 s window at 256 Hz over 19 channels of the 10–20 montage, robust-normalized per recording,  $x \in \mathbb{R}^{B \times 2560 \times 19}$ .

**Filterbank tokenizer.** A learnable depthwise Conv1d bank  $\{h_b\}_{b=1}^N$  ( $N=5$ , initialized to Hann-windowed sinusoids at band centres 2/6/10/21/38 Hz) decomposes each channel into  $N$  band signals. Each 200-sample patch of each band is linearly embedded; the  $N$  band embeddings are combined and projected to the token dimension  $D=512$ . The tokens  $[B, C, T_p, D]$  ( $C=19$ ,  $T_p=12$ ) are processed by a 12-layer criss-cross transformer that alternates channel and time attention (Fig. 2).

**Spectral prediction (primary).** We mask a random fraction of *time* patches; masked patch embeddings are replaced by a learnable token so the encoder never sees them. From the encoder output we predict the log band-power of each masked patch and supervise it against a target computed by a *fixed* (buffer, no-gradient) filterbank  $h^{\text{fix}}$  on the raw signal:

$$\mathcal{L}_{\text{cf}} = \frac{1}{|\mathcal{M}|} \sum_{(c,t,b) \in \mathcal{M}} (\hat{p}_{c,t,b} - \log(1 + \bar{e}_{c,t,b}))^2, \quad (1)$$

where the target band power is  $\bar{e}_{c,t,b} = \frac{1}{P} \sum_{\tau} (h_b^{\text{fix}} * x)_{c,t,\tau}^2$  and  $\mathcal{M}$  is the set of masked (channel, patch, band) triples. A fixed target is critical: with a *learnable* target filterbank the model minimizes  $\mathcal{L}_{\text{cf}}$  by collapsing the filters so the target

Table 1: **Main results (balanced accuracy).** Leak-free benchmarks. Baseline rows as reported by their authors (each pretrained on more or comparable data: CBraMod/CSBrain  $\sim 9,000$  h, LaBraM  $\sim 2,500$  h). Ours is pretrained on only  $\sim 2,500$  h. Mumtaz/MA led by frozen probe (deterministic); see Table 3.

|                  | ISRUC                               | HMC                                 | Mumtaz            | MentalArith       |
|------------------|-------------------------------------|-------------------------------------|-------------------|-------------------|
| LaBraM           | –                                   | –                                   | 0.941             | –                 |
| CBraMod          | 0.787                               | –                                   | 0.956             | 0.726             |
| CSBrain          | 0.793                               | 0.735                               | 0.964             | 0.756             |
| Random (FT)      | $0.738 \pm 0.008$                   | ?                                   | ?                 | ?                 |
| <b>Ours (FT)</b> | <b><math>0.793 \pm 0.005</math></b> | <b><math>0.748 \pm 0.005</math></b> | $0.913 \pm 0.026$ | $0.763 \pm 0.073$ |

$\rightarrow 0$  (we observe  $\mathcal{L}_{\text{cf}} \rightarrow 4 \times 10^{-4}$  and degenerate features). A fixed target makes collapse counter-productive: an uninformative encoding cannot predict real band energy, which anchors the encoder to stay informative.

**Auxiliaries.** A light raw-signal reconstruction term (weight 0.5) regularizes the decoder, and SIGReg (Balestriero and LeCun 2025) enforces isotropy (weight  $\lambda=0.05$ ). We use *no* latent JEPA term (an ablation found it inert here):  $\mathcal{L} = \mathcal{L}_{\text{cf}} + 0.5 \mathcal{L}_{\text{mae}} + 0.05 \mathcal{L}_{\text{sig}}$ .

## Experiments

**Pretraining.** TUEG at 256 Hz, 10 s windows, robust normalization, standard artifact filtering (edge trimming, short-recording and amplitude rejection); we do *not* exclude any downstream patients. Our primary model uses a random  $\sim 2,500$ -hour subset (2,100 patients drawn by a fixed seed); a matched-compute control uses the full  $\sim 17,666$  hours (14,764 patients) with an identical gradient budget (same effective batch,  $7 \times$  fewer epochs).

**Benchmarks (all leak-free).** ISRUC, HMC: 5-class sleep staging (sequence-to-sequence). Mumtaz2016: depression (binary). Mental Arithmetic (eegmat): rest vs. arithmetic (binary). We use the CBraMod/CSBrain subject-disjoint splits.

**Protocol and checkpoint policy.** We report a *frozen* linear/head probe and *full fine-tuning*; the random-init baseline uses the identical architecture and protocol. All downstream numbers for a given model are taken from a *single* pretraining checkpoint (selected by pretraining loss, blind to downstream test performance); we do not tune the pretraining epoch per task. Because Mumtaz (11 test subjects) and Mental Arithmetic (4 test subjects) test sets are tiny, fine-tuning is high-variance on them; we therefore lead with the deterministic frozen probe and report fine-tuning as mean $\pm$ std over 5 seeds. Sleep sets are large; we report 3 reps.

## Results

*Draft: several cells are pending final-checkpoint / auxiliary runs and are marked ?; sleep numbers are from the current best checkpoint of the  $\sim 2,500$ -hour model and will be fixed to the selected checkpoint.*

**Sleep staging at one-seventh the data.** Our  $\sim 2,500$ -hour model reaches ISRUC balanced accuracy  $0.793 \pm 0.005$  and HMC  $0.748 \pm 0.005$  (Table 1), matching CSBrain on ISRUC

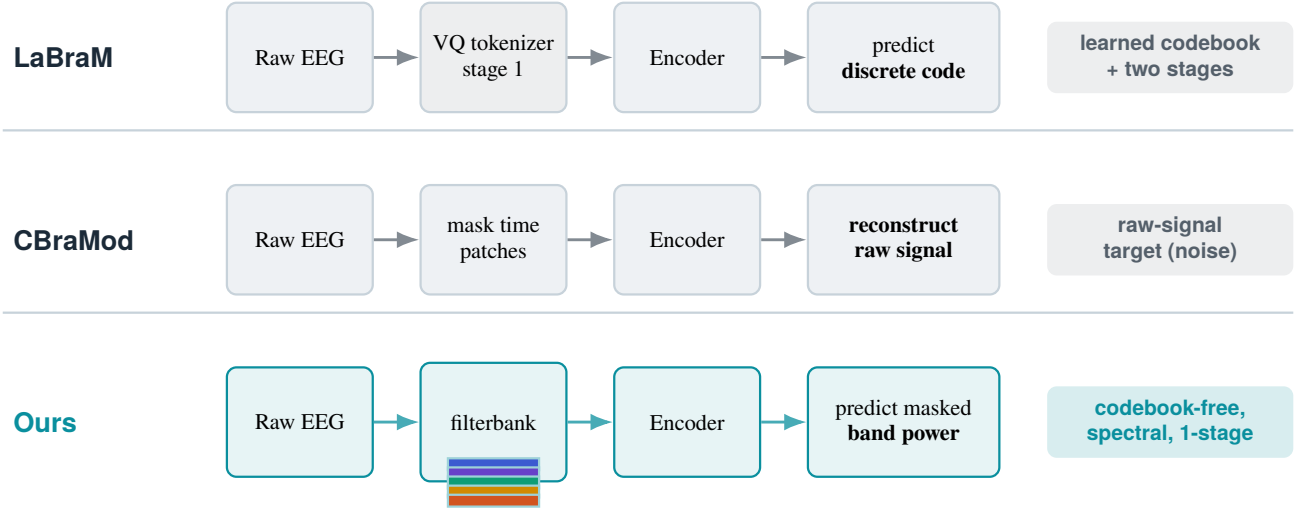


Figure 1: **Codebook-free spectral prediction vs. prior EEG foundation models.** LaBraM predicts discrete codes from a separately trained VQ tokenizer; CBraMod reconstructs the raw masked signal. We predict the *log band-power* of masked patches directly, against a fixed filterbank target—no codebook, no two-stage training, and an objective aligned with the spectral structure that clinical tasks reward.

Table 2: **Data efficiency (ISRUC / HMC balanced accuracy).** Our  $\sim 2,500$ -hour model vs. a matched-compute control on the full corpus and vs. prior methods with their pre-training budgets. Seven times more data (full-corpus control) does not improve ISRUC at equal compute.

| Model            | Pretrain data   | ISRUC                               | HMC                                 |
|------------------|-----------------|-------------------------------------|-------------------------------------|
| LaBraM           | $\sim 2,500$ h  | —                                   | —                                   |
| CBraMod          | $\sim 9,000$ h  | 0.787                               | —                                   |
| CSBrain          | $\sim 9,000$ h  | 0.793                               | 0.735                               |
| Ours (full ctrl) | $\sim 17,666$ h | 0.798                               | ?                                   |
| <b>Ours</b>      | $\sim 2,500$ h  | <b><math>0.793 \pm 0.005</math></b> | <b><math>0.748 \pm 0.005</math></b> |

(0.793) and *exceeding* it on HMC (0.735;  $+0.014$ ,  $\approx 2.8\sigma$ ), while CBraMod and CSBrain pretrain on  $\sim 9,000$  hours. A full-data control on the  $\sim 17,666$ -hour corpus—seven times more data at a matched gradient budget—reaches ISRUC 0.798, statistically indistinguishable from the  $\sim 2,500$ -hour model (Table 2). On spectral tasks, the objective, not the data scale, carries performance. On the tiny-cohort clinical tasks the model is competitive (Mental Arithmetic 0.763, above CSBrain’s 0.756) but high-variance under fine-tuning, so we lead with the deterministic frozen probe there (Table 3).

**Checkpoint selection.** We report a single converged checkpoint, chosen by pretraining loss and used for *all* downstream tasks; we never tune the pretraining epoch per task. It sits at roughly the matched gradient budget of the full-corpus control ( $\sim 7\times$  more epochs on  $\sim 7\times$  less data). Earlier checkpoints underfit; much later ones overfit the small subset and degrade the tiny-cohort clinical tasks (e.g. Mental Arithmetic falls from 0.763 to 0.617), so the converged point is both the

Table 3: **Frozen linear probe** (deterministic). Pretrained features vs. random init on the tiny-cohort clinical tasks, where we lead with the frozen probe to avoid fine-tuning variance.

|             | Mumtaz BA | MentalArith BA |
|-------------|-----------|----------------|
| Random      | ?         | ?              |
| <b>Ours</b> | ?         | <b>0.677</b>   |

Table 4: **Objective ablation** at  $\sim 2,500$  h, matched compute (same subset, same steps; only the loss changes). Isolates spectral band-power prediction (CF) vs. raw-signal reconstruction (MAE).

| Objective                     | ISRUC                               | HMC                                 | Mumtaz            |
|-------------------------------|-------------------------------------|-------------------------------------|-------------------|
| MAE only (reconstruction)     | ?                                   | ?                                   | ?                 |
| CF only (spectral prediction) | ?                                   | ?                                   | ?                 |
| <b>CF + MAE (ours)</b>        | <b><math>0.793 \pm 0.005</math></b> | <b><math>0.748 \pm 0.005</math></b> | $0.913 \pm 0.026$ |

principled comparison to the control and the natural operating point.

**Label efficiency.** A frozen linear probe on our features approaches fine-tuned accuracy with a small fraction of downstream labels (Fig. 3): on Mumtaz, ?% of labels already reach within ? BA of the full-label probe, whereas a random encoder needs all labels to reach far lower accuracy.

## Ablations

**Spectral target vs. signal reconstruction.** Table 4 compares, at identical data and compute, our band-power target (CF) against pure raw-signal reconstruction (MAE) on the same backbone. [Result pending: if CF-only  $\approx$  full > MAE-

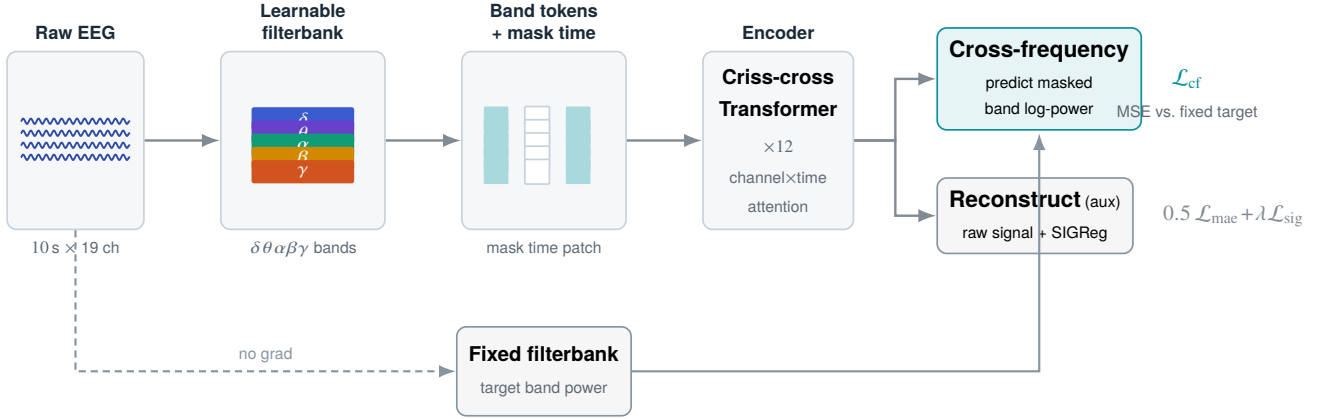


Figure 2: **Codebook-free spectral-prediction pretraining.** A learnable filterbank tokenizer splits each patch into  $\delta\theta\alpha\beta\gamma$  bands; time patches are masked out of the encoder input. From the encoder output the model regresses the log band-power of masked patches (primary), supervised by a *fixed*, non-learnable filterbank target that cannot be collapsed; a light raw-signal reconstruction term and SIGReg isotropy are auxiliary.

*only, spectral prediction is the load-bearing ingredient and reconstruction is unnecessary.]*

**Fixed vs. learnable target.** A learnable-target filterbank collapses ( $\mathcal{L}_{cf} \rightarrow 0$ , degenerate features); the fixed target is necessary and sufficient to prevent this.

### Limitations

Our model is designed for the spectral, quasi-stationary structure of clinical and passive EEG. It is *not* suited to motor imagery, whose discriminative signal is a spatially localized event-related (de)synchronization rather than a stationary band-power pattern; we exclude motor-imagery benchmarks by design rather than reporting weak numbers. Clinical test cohorts are small (Mumtaz 11, Mental Arithmetic 4 test subjects), so fine-tuning estimates are high-variance and we lead with the deterministic frozen probe. We also exclude TUH-derived abnormality detection (TUAB) from the main comparison: it overlaps at the patient level with TUEG pretraining for every method, and is off-topic for our spectral scope.

### Conclusion

Predicting band power—directly, against a fixed filterbank target, with no codebook and no two-stage tokenizer—is a simple objective that is well matched to the spectral structure of clinical EEG. It reaches state-of-the-art sleep staging with a fraction of the pretraining data used by prior foundation models, is label-efficient downstream, and makes the surprising point that for spectral EEG tasks the pretext *objective*, not the data scale, is what matters. We argue this reframes what an EEG foundation model should predict.

### References

Balestriero, R.; and LeCun, Y. 2025. LeJEPa: sketched isotropic Gaussian regularization for joint-embedding predictive architectures.

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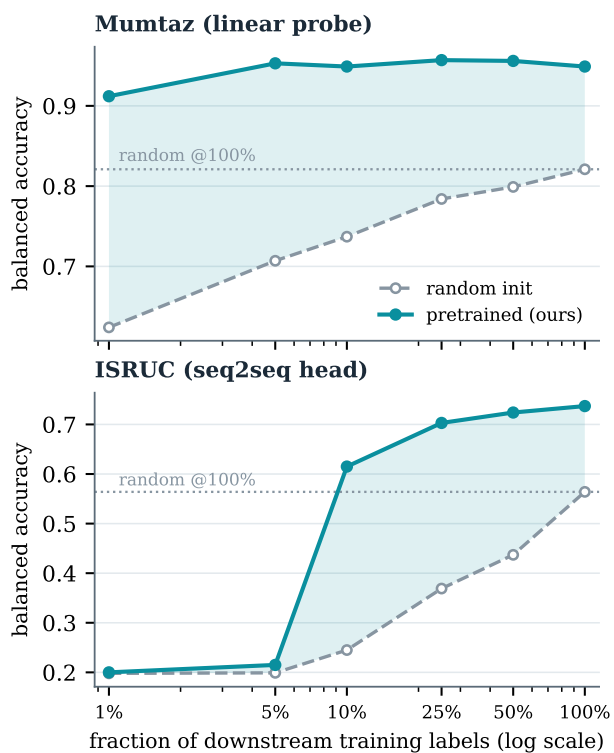


Figure 3: **Label efficiency.** Balanced accuracy of a frozen linear probe as a function of the fraction of downstream training labels, pretrained (ours) vs. random init. Pretrained features are accurate at 1–10% labels; the two converge only near 100%.